Interdomain Collaboration Between Hyperspectral and VHR Remote Sensing Images: A Cross-Scene Few-Shot Learning Framework for Change Detection

Xianghai Wang, Siyao Li, Xiaoyang Zhao, and Yuetong Zhao

Abstract— **Hyperspectral image change detection (HSI-CD) based on deep learning (DL) has made significant progress. However, these methods rely significantly on the number of labeled data. Annotating HSI is a highly complex task that requires professional knowledge for guidance, resulting in a scarcity of high-quality labeled samples. The emergence of few-shot learning (FSL), which supports model learning from limited labeled samples, can address this issue. However, FSL-based methods still face some challenges: 1) existing methods mainly rely on single-source or homogenous cross-domain HSI data, which is difficult to adequately cope with the problem of scarcity of HSI labeled data; 2) most existing methods usually only focus on local features within patches and neglect interrelationships between patches, which is also important for model learning; and 3) transformers modeling long-range relationships rely on extensive labeled data, making it difficult to perform well in few-shot scenarios. Therefore we propose a cross-scene FSL framework based on interdomain collaboration (CSIDC-FSL) for HSI-CD. Specifically, the following is proposed: 1) FSL is performed on very high-resolution image (VHRI) and HSI, aiming to use the learnable information in VHRI with low annotation cost to help HSI-CD, reducing the dependence of the model on HSI annotation data while enabling multilevel feature hybrid perceptual CD; 2) a dual-information integrated mapping module (DI2M) is proposed, which designs a CNN and transformer integrated structure that can simultaneously focus on local features and class-wise long-range relationships to break the constraints of local perception of CNN while optimizing the performance of transformer under few-shot situations; and 3) the interdomain joint information allocation module (IDM) is designed to capture cross-scene domain-wise distribution features, and mitigate the impact of distribution differences in cross-scene data (VHRI and HSI) on knowledge learning and migration through the collaboratively consistent interdomain features. Under the condition of five samples per class, the CD results of CSIDC-FSL are better than those of recently advanced algorithms, with average improvements of 1.46%–1.5% for overall accuracy (OA) and average accuracy (AA), respectively. The code will be made available at https://github.com/lsylnnu/CSIDC-FSL.**

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I. Introduction

THE high spectral resolution of hyperspectral images (HSIs) is capable of capturing hundreds of nearly continuous spectral bands. This abundant spectral information has unique advantages in fine-grained observations of the Earth [1], [2]. Among them, HSI change detection (CD) is a technology used to discover surface changes by analyzing HSIs of the same surface area taken at different times. It is widely used in various industries, such as urban development design and management, crop growth and yield detection, and monitoring of environmental changes [3], [4].

In the initial development stage, HSI change detection (HSI-CD) mainly includes principal component analysis (PCA)-based methods for CD by analyzing the principal component differences of HSIs in different phases [5], Markov random field model-based CD methods that classify difference images [6], and change vector analysis (CVA)-based methods identify changes by comparing the direction and intensity of feature vectors [7]. However, most of these methods divide the changed features by simple linear mapping and threshold calculation. They usually rely on manual extraction of a large number of features, which heavily relies on empirical values and the process that is prone to introduce artificial noise that interferes with the accuracy of the final detection.

With the rise in deep learning (DL) techniques, many DL methods have emerged that can more accurately and automatically capture and describe features. Convolutional neural network (CNN), as a representative network structure within DL, has been valued by the field of HSI-CD due to its good feature mining ability and stable representation learning ability [8]. For example, Wang et al. [9] combined the attention mechanism with a Siamese CNN and proposed a network named SSA-SiamNet, which could focus on more discriminative features while suppressing redundant information to thoroughly utilize the joint spatial–spectral information in HSI; Zhao et al. [10] designed a 3-D CNN autoencoder (S3DCAECD), engineered to extract both spatial and spectral characteristics. The features, thus extracted, were subsequently

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fed into a trained classifier to facilitate CD; Wang et al. [11] proposed an HSI-CD framework that could incorporate subpixel contextual information for feature learning, i.e., subpixel convolution-based residual network (SPCNet). These methods utilize CNN to mine changed information and achieve better CD results. However, they usually use HSI patches as the input of the network, and perform local feature extraction on each patch through convolution operation, without being able to learn the global information of HSI as a whole. In fact, paying attention to the correlation between patches helps the model understand image information with a larger receptive field and to learn more appropriate and accurate changed characteristics. Therefore, long-range information has a positive effect on CD. To better realize CD, this long-range information should be used reasonably [12].

Nowadays, the transformer, which can model long-range dependencies, provides an effective solution to the above problem [13]. For example, Wang et al. [14] focused on the importance of HSI temporal information and proposed a triplet transformer framework (TriTF). Difference-temporal image information was introduced, and a novel module of cross-temporal attention mechanism was constructed, so the feature representation of the network was enhanced; Ding et al. [15] introduced the transformer-encoder into HSI-CD and proposed a CDFormer network to mine the correlation between bitemporal features while capturing more discriminative changed features through the weighted union of spatial–spectral–temporal features. Although all of the above methods have achieved decent results, they often rely on a large number of labeled samples to train the model or classifier to have good feature mining and learning capabilities. Specifically, the transformer differs from CNN in that it lacks the local inductive bias inherent in CNN, and its ability to comprehend global information requires a substantial number of parameters to fit. This also means that the transformer requires more labeled samples to participate in training, and its performance is not even as good as the CNN-based HSI-CD methods under the same few-shot conditions [16], [17]. But in fact, the production of labeled data required by the HSI-CD task is very difficult and sometimes even impossible [18]. For example, complex geographic areas such as coastal wetlands will not be able to obtain a large number of manually labeled samples due to circumstances including the large scope of the area and the inaccessibility.

To address the issue of having an inadequate number of HSI labeled data, there are various solutions such as unsupervised learning [19], semisupervised learning [20], domain adaptation strategy [21], few-shot learning (FSL) [22], and other methods. Among them, the domain adaptation strategy aims to reduce the feature differences between different domains, so that the model trained on the source domain with sufficient labels can be adaptively applied to the target domain with insufficient labels. For example, Qu et al. [23] designed a network based on a graph domain adaptation strategy that could fully utilize the feature information in both source and target domains, enhancing the generalization capability of the model and ensuring good performance at small-sample-size sampling rates; Qu et al. [24] proposed an unsupervised HSI-CD method based on domain adaptation. A cycle strategy was designed to achieve the goal of learning changes on the source domain and applying the model to the target domain by gradually mitigating domain distribution differences and updating to obtain high-confidence training samples. Typically, researchers combine domain adaptation strategy with semisupervised or unsupervised learning to design models. Although this alleviates the problem of insufficient labeled data to some extent, model performance still depends on small-sample-size samples (such as 100–200 labels) or is contingent on the quality of pseudo labels. In practical tasks, there are often only very few number of labels available (such as 5–10 labels). At this point, the performance of models based on domain adaptation strategy needs to be further improved. For such cases, models based on FSL often perform better. FSL is a learning strategy that aims to enable the model to complete feature learning and ultimately recognize both learned and new features using finitely labeled data. Compared to other solutions, the FSL-based model can learn the difference between different classes of features and build awareness of new features with only a few training samples. This alleviates the problem of lack of sufficient training samples for HSI and helps the model to acquire fast learning capability and excellent generalization ability. Nowadays, due to the above advantages, FSL has received more and more attention, e.g., Liu et al. [25] combined FSL and DL to construct a framework called DFSL, which extracted HSI spatial–spectral features by deep residual CNN and constructed a classifier by applying FSL metric on the features to achieve classification; Zhang et al. [26] put forward a novel method combing domain adaptation with FSL (DAFSL), which first performed embedded feature extraction and domain adaptation through the encoder, and subsequently performed FSL on the features of the source domain and target domain together and fine-tuned the classifier to accomplish few-shot classification; Liu et al. [27] constructed a hybrid relational network (H-RNet) based on FSL to cope with the challenge of limited labeled data. The discriminative spatial–spectral features were obtained by the hybrid network, while FSL was utilized to alleviate the issue of insufficient labeled samples. Although the above FSL-based methods have greatly reduced the number of labeled samples required for training, most FSL methods still use single-source data or cross-domain data from the same source. The single-source HSI data lacks labeled samples and is prone to overfitting even though FSL is performed. Moreover, the HSI annotation is both time-consuming and laborious. When both the source and the target domains utilize HSI, it leads to higher labeling costs without effectively addressing the issue of limited labeled samples in HSI. In reality, many other types of data have sufficient labeled samples, and the labeling cost is lower than that of HSI, such as very high-resolution image (VHRI) data [28], [29]. Therefore, these annotation data should be effectively utilized to assist HSI with insufficient labeled samples to complete CD.

Based on the above problems, a cross-scene FSL framework based on interdomain collaboration (CSIDC-FSL) is proposed, aiming to build an FSL-based model with complementary advantages of CNN and transformer, using limited labeled data

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to effectively mine local information and explore long-range dependencies to better complete CD. The main contributions are listed below.

1) Proposing an FSL framework for cross-scene HSI-CD, named CSIDC-FSL. In this framework, FSL is performed separately on VHRI and HSI data to migrate learnable information and decrease reliance on labeled HSI data in training. Additionally, CSIDC-FSL enables multilevel feature hybrid perceptual CD of patch-wise, class-wise, and domain-wise.
2) A dual-information integrated mapping module (DI2M) is proposed, which fully pays attention to local features and class-wise long-range dependencies to break the constraints of CNN local perception. The designed integrated structure can introduce appropriate inductive bias to the transformer, enabling it to perform well despite the limited training samples.
3) An interdomain joint information allocation module (IDM) is designed to capture the intradomain personality distribution features and interdomain commonality distribution features of the source and target domains, and coordinate the personality features by allocating the commonality features into intradomain. Ultimately, distributions are aligned to make the distributions of the two domains through collaborative consistency to mitigate the impact of the inherent feature distribution differences of cross-scene data on FSL.

The rest of this article is structured in the following way: to aid comprehension of the article, some necessary relevant information will be provided in Section II, including knowledge of cross-scene FSL, transformer, and domain adaptation. In Section III, the proposed framework is described in terms of its overall flow, specific modules, loss function, and testing process. The experiment-related content will be presented in Section IV. Section V is the discussion section and Section VI is the final conclusion section.

II. Related Work

A. Cross-Scene FSL

FSL is a learning method that can learn quickly using fewer training samples, and it aims to learn the differences between features of different classes of objects, which enables the model to have the ability to classify new features that have not been seen before [30], [31]. This ability of FSL makes it widely used in cross-scene tasks in multiple fields. Specifically, cross-scene FSL is dedicated to extracting learnable features from the source domain (dataset with sufficient labeled) and transferring them to the target domain (dataset with insufficient labeled), helping the target domain to accomplish corresponding tasks, while utilizing the FSL to reduce the amount of labeled data required for training and to better realize the knowledge transfer between cross-scene data. Nowadays, many cross-scene FSL-based classification methods have been proposed. Li et al. [32] proposed a cross-domain DL method (DCFSL), which combined cross-scene FSL for the HSI classification task at the first time; Zhang et al. [33] combined cross-scene FSL with contrast learning to learn more discriminative representations and improved classification accuracy; Liang et al. [34] focused on the situation where the labeled data in the target domain was limited but there were available labels in the source domain that could be used for training, and therefore designed an attention multisource fusion cross-scene FSL method (AMF-FSL) to address the restriction of training due to the insufficient labeled samples of the single-source data and to achieve HSI classification under few-shot conditions.

In short, most current research based on cross-scene FSL still focuses on solving the classification problems. As for the CD task, there is a large intraclass difference among the features of the changed class, and it is more challenging for the model to learn the features of the changed class under the condition of few shot, so how to realize the capture of the real changed features and the identification of the pseudo changed features, enabling FSL to utilize accurate features for learning and be suitable for CD tasks is still a new subject. Moreover, most of the existing methods use the same type of data source, i.e., either source or target domains use HSI. However, there are general difficulties in annotating HSI data. It is worth exploring whether other types of data with lower annotation costs can be used as the source domain (such as VHRI) to assist in implementing various HSI processing tasks.

B. Transformer in HSI Processing

The transformer, a pioneering new network architecture, is originally proposed in the field of natural language processing to implement machine translation tasks. In the field of computer vision, the transformer learns contextual information by modeling the affinity between any two pixels through the self-attention mechanism [35]. Because of its powerful long-range modeling capability, the transformer has found extensive use in various other fields, including HSI processing. For example, in the field of HSI classification, He et al. [36] introduced the bidirectional encoder structure of transformer (BERT) and proposed a classification framework with a global receptive field, named HSI-BERT, which utilized the multihead self-attention mechanism in the BERT model to perceive the global information and obtain more discriminative features; Zhou et al. [37] proposed an HSI classification model that combined recurrent neural network (RNN) with transformer, which utilized RNN to construct pixel sequences and assign appropriate positional attention weight to pixels in the sequences via transformer to generate sequence-level features; Yu et al. [38], aiming to solve the problem of limited receptive field of current CNN-based methods, proposed a lightweight transformer-based model (MSTNet). Through the designed encoder–decoder, multilevel feature learning and fusion are performed on the processed HSI sequences, and finally realized the classification; in the field of HSI reconstruction, Cai et al. [39] focused on the fact that modeling long-range dependencies between HSI spectra could help in better implementation of HSI reconstruction tasks, and thus proposed a mask-guided spectral intelligence transformer (MST) to explore the HSI interspectra interactions; furthermore, in the field of HSI-CD, Wang et al. [40] considered the

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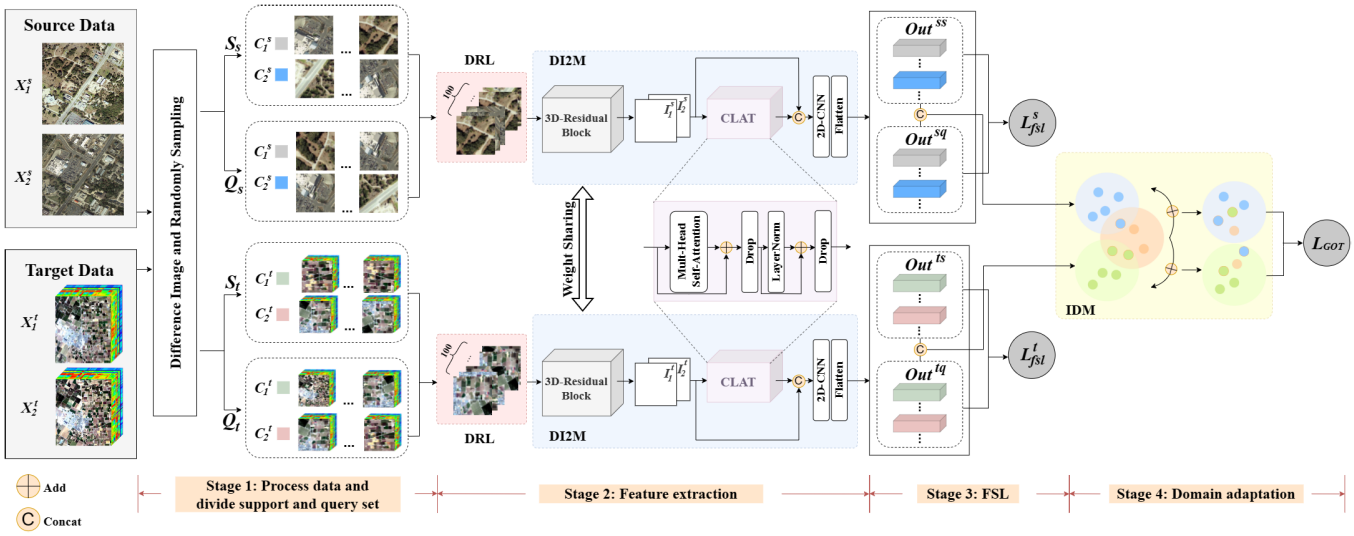

Fig. 1. Training process of the proposed CSIDC-FSL framework includes some important stages: feature extraction, FSL, and domain adaptation. For the data processing, the bitemporal patches of the VHRI and HSI are correspondingly subtracted to obtain the corresponding difference patches of the VHRI and HSI, and then random sampling is performed to obtain the $\{S_s,\ Q_s,\ S_t,\ Q_t\}$ and start training. First, the feature dimensions are unified by a layer of 2-D convolutional layer in DRL for the subsequent stages. Subsequently, the extraction and integration of local features and class-wise long-range dependency features are performed in the feature extraction stage using the DI2M indicated by the blue rectangles in the figure. In the FSL stage, a distance metric is applied to the extracted features. This helps to differentiate between changed and unchanged classes more efficiently when labeling information is limited. Ultimately, domain adaptation using the IDM, indicated by the yellow rectangles in the figure, reduces the negative impact of the differences in feature distribution caused by different scene data on the model learning performance.

importance of sequence information on HSI and proposed a transformer framework (SST-Former) for HSI-CD to extract spatial–spectral–temporal sequence information.

Overall, the transformer has been used flexibly in various tasks related to HSI processing and has shown positive outcomes. However, it lacks the inductive bias that CNN has. Compared to CNN, the transformer requires more labeled samples to train numerous parameters for better global perception and feature attention ability. However, the problem of the difficulty of HSI annotation cannot be ignored. It is worth exploring ways to maintain the high performance of the transformer in few-shot conditions and make it suitable for the HSI-CD framework.

C. Domain Adaptation

In the cross-scene HSI task, the source domain and the target domain will inevitably have different interdomain feature distributions due to the differences in the surface scenes, which will negatively affect the effectiveness of interdomain knowledge transfer, i.e., there is a domain shift problem [41], [42]. Domain adaptation is a method that can effectively solve this problem.

Huang et al. [43] proposed a graph embedding-based domain adaptation (GEDA) method, which minimized the interdomain distribution differences by performing distribution alignment and subspace alignment on the spatial–spectral discrimination features of two domains; Peng et al. [44] proposed a domain adaptation method for matching source and target features in the kernel PCA space by minimizing the empirical maximum mean difference; Huang et al. [45] proposed a new method of domain adaptation that could effectively utilize detailed information in both spatial and spectral dimensions, i.e., the two-branch attention adversarial domain adaptation (TAADA) method, which was mainly based on the mutual adversarial learning between the generator and the discriminator, to continually adjust the distributions between the source domain and the target domain to complete the interdomain feature distribution alignment.

From the above methods, most of the domain adaptation methods usually focus only on the extraction of intradomain discriminative distribution features, ignoring the interdomain commonality distribution features, while experiments show that the learning of interdomain dependencies has a positive guiding effect on the realization of domain adaptation [46]. Therefore, how to effectively combine and utilize the intradomain information and interdomain information of two domains to better achieve domain adaptation is a problem of particular importance and worth researching.

III. Methodology

In this section, the input data processing and training flow of the proposed CSIDC-FSL will be described first. Then the modules in the framework will be described separately, including two innovative modules and the implementation of the cross-scene FSL. The loss function is then defined and finally is the testing phase. Fig. 1 is the structure of the CSIDC-FSL.

A. Overall Process of CSIDC-FSL

The input of CSIDC-FSL consists of source domain data and target domain data. In the experiment, we used VHR bi-temporal images with sufficient labeled samples as source domain data, and HSI bitemporal images as target domain data. First, the difference VHRI is obtained by directly subtracting the bitemporal images. Similarly, the bitemporal images of the target domain are done in the same operation

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to obtain the difference HSI. Then, the obtained difference images are sliced into patches. Define the dataset consisting of the obtained source domain image patches as $X_s = \{x_i^s | i = 1, \ldots, N\}$, $X_s \in \mathbb{R}^{p\times p\times chs}$, where x_i^s is the ith source domain image patch, N is the number of source domain image patches, $p \times p$ is the spatial dimension size of the image patches, and chs is the number of spectral dimension bands of the source domain image patches. Similarly, the obtained target domain image patches dataset is $X_t = \{x_j^t | j = 1, \ldots, M\}$, $X_t \in \mathbb{R}^{p\times p\times cht}$, where x_j^t is the jth target domain image patch, M is the number of target domain image patches, and cht is the number of spectral dimension bands of the target domain image patches. We randomly select 500 labeled samples in X_s and five labeled samples in X_t for training, and add Gaussian noise to the samples in X_t to enrich the training samples. Subsequently, to facilitate cross-scene FSL, we first divide support and query sets. Specifically, W samples per class are selected from the training samples of the source domain to form the source domain support set $S_s = \{(x_w^s, y_w^s)\}_{w=1}^{C\times W}$, where x_w^s is the wth sample, y_w^s is its corresponding label, and C is the number of classes ($C = 2$, i.e., changed class and unchanged class), noting that C_1^s is the first class of the source domain and C_1^t is the first class of the target domain. Subsequently, K samples are selected from the remaining samples of per class to form the source domain query set $Q_s = \{(x_k^s, y_k^s)\}_{k=1}^{C\times K}$, where x_k^s is the kth sample and y_k^s is its corresponding label. In the same way, the target domain support set $S_t = \{(x_w^t, y_w^t)\}_{w=1}^{C\times W}$ and the query set $Q_t = \{(x_k^t, y_k^t)\}_{k=1}^{C\times K}$ are obtained. Finally, the samples in S_s, Q_s, S_t, and Q_t are input into CSIDC-FSL for training.

The training process mainly includes feature extraction, FSL, and domain adaptation.

1) First, the spectral dimensions of the VHRI and HSI are unified through the dimensionality reduction layer (DRL). Afterward, patch-wise and class-wise features are captured and integrated through DI2M, and then the features are mapped to an embedding space (refer Section III-B).
2) Subsequently, enter the FSL stage. The support set and query set features of source and target domains in the embedding space are measured to complete the cross-scene FSL (refer Section III-C).
3) Finally, domain adaptation is performed. The features of the two domains are further mined by IDM, allocating the captured interdomain commonality features to combine them with the personality features. Afterward, the distributions of the two domains are coordinated consistently by feature distribution alignment (refer Section III-D).

B. Dual-Information Integrated Mapping Module

For the convenience of subsequent processing, the DRL consisting of a plain 2-D convolutional layer is first used to extract features from VHRI and HSI, and their feature dimensions are unified to map to F (F is set to 100 in the experiment), that is, $\{S_s, Q_s, S_t, Q_t\} \in \mathbb{R}^{p\times p\times F}$. Subsequently, they are further processed by DI2M.

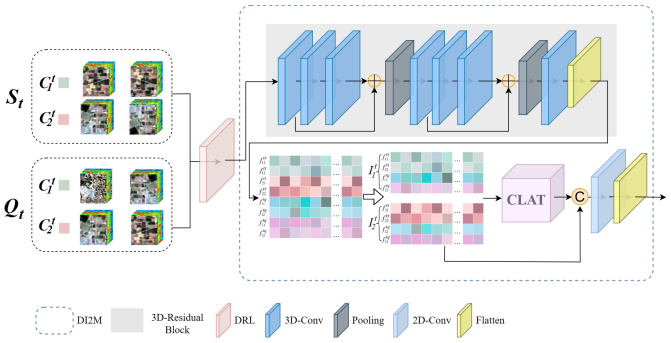

Fig. 2. Dual-information integrated mapping module.

We notice that the traditional CNN-based HSI-CD frameworks can only focus on local feature within the fixed-size patch of the input when extracting feature, and are unable to perceive the global information of the image. Although the good long-range modeling capability of the transformer is currently used in several fields, the framework consisting solely of the transformer is powerless for the few-shot problem caused by its lack of inductive bias. Therefore, we design the following DI2M.

First, the local feature is extracted from each input patch by a 3-D residual block consisting of multiple 3-D convolutional (3D-Conv) layers, and the specific connection between the convolutional layers in the block is shown in Fig. 2. The extracted local feature can be expressed as follows (taking one batch of target domain input data as an example, the same below):

$$\begin{cases} f^{\mathrm{ts}} = \mathrm{RS}\left(x^{\mathrm{ts}}\right) \\ f^{\mathrm{tq}} = \mathrm{RS}\left(x^{\mathrm{tq}}\right) \end{cases} \tag{1}$$

where f^{ts} is the local feature of one patch in the S_t, f^{tq} is the local feature of one patch in the Q_t, $\mathrm{RS}(\cdot)$ is the 3-D residual block, x^{ts} is a sample (one patch) in S_t, and x^{tq} is a sample in Q_t.

Subsequently, a class-wise attention transformer (CLAT) is constructed, which is a transformer-based variant structure. The designed layers and their subtle connections enable CLAT to adapt to the overall architecture and achieve better results. CLAT mainly utilizes the multihead self-attention mechanism in the transformer to mine the relationships between the samples in the support set and the query set. Specifically, we focus on the relationships between the various classes of support set samples and the different query set samples, and the capture of such relationships can assist in the subsequent execution of the FSL. The process is shown in Fig. 2, which takes the example that the target domain data has two classes, and two samples per class are selected to form S_t and Q_t as the input data of DI2M. First, based on the local features f^{ts} and f^{tq}, which are extracted by 3-D residual block in the early stage, it is defined that f_{11}^{ts} is the local feature of the first sample of class 1 in S_t, f_{21}^{ts} is the local feature of the first sample of class 2 in S_t, f_{12}^{tq} is the local feature of the second sample of the class 1 in Q_t, and the rest is the same. Afterward, the local features of the same

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class in S_t are combined with a portion of randomly selected local features in Q_t. Obtain I_1^t as the local feature sequence of class 1 of S_t, and I_2^t as the local feature sequence of class 2 of S_t. The class-wise local feature sequences for different classes of S_t can be expressed as follows:

$$I_c^t \leftarrow \mathrm{CAT}\left(f_{c1}^{\mathrm{ts}}, \ldots, f_{cz}^{\mathrm{ts}}, f_1^{\mathrm{tq}}, \ldots, f_h^{\mathrm{tq}}\right)\ (c = 1, 2) \quad (2)$$

where CAT is the connection operation, c is the number of classes, $c1$ denotes the first sample in class c, and cz denotes the zth sample in class c. f_h^{tq} is the hth f^{tq}, and h is the number of samples randomly selected from the 1 batch input of Q_t. I_c^t is the class-wise local feature sequence of the cth class, which is obtained by performing the connection operation between the f^{ts} of the cth class and h randomly selected f^{tq} in 1 batch.

The class-wise local feature sequences are sequentially input into CLAT; first, the queries, keys, and values adapted to the dimension of the number of heads in the multihead self-attention are obtained by linear mapping as well as mapping function processing, denoted as Q_c^t, K_c^t, V_c^t, where Q_c^t denotes the query obtained by I_c^t and the rest is the same. Next, the dot product operation and Softmax function are applied between Q_c^t and K_c^t to obtain the class-wise correlation score matrix, which is then dot produced with V_c^t. Finally, the dimension of the matrix is rearranged and mapped by linear mapping to obtain an output sequence of the same dimension as the input. The above process can be represented as follows:

$$\begin{cases} Q_c^t, K_c^t, V_c^t = \mathrm{map}\left(\mathrm{Lin}\left(I_c^t\right)\right) \\ \mathrm{Att}_{ts\leftrightarrow tq}\left(Q_c^t, K_c^t\right) = \mathrm{Softmax}\left(\dfrac{Q_c^t\left(K_c^t\right)^T}{\sqrt{d}}\right) \\ \mathrm{Out}_c^{t0} = \mathrm{Drop}\left(\mathrm{Lin}\left(\mathrm{Att}_{ts\leftrightarrow tq}\left(Q_c^t, K_c^t\right)V_c^t\right)\right) \end{cases} \quad (3)$$

where map is the mapping function, $\mathrm{Lin}(\cdot)$ is the linear layer, $\mathrm{Att}_{ts\leftrightarrow tq}(Q_c^t, K_c^t)$ is the class-wise correlation score matrix, Softmax is the activation function, $(K_c^t)^T$ denotes the transpose matrix of K_c^t, d denotes the feature dimension of K_c^t, the final output class-wise correlation feature sequence of the cth class is denoted by Out_c^{t0}, and Drop is the drop operation.

Subsequently, Out_c^{t0} is residually concatenated with layer normalization layer and drop layer, and the result is integrated with the local features captured by the 3-D residual block, and finally, the final integrated feature sequence for each class is output. The dual information integration process is as follows:

$$\begin{cases} \mathrm{Out}_c^{t1} = \mathrm{Drop}\left(\mathrm{Out}_c^{t0} + I_c^t\right) \\ \mathrm{Out}_c^{t2} = \mathrm{Drop}\left(\mathrm{LN}\left(\mathrm{Out}_c^{t1}\right) + \mathrm{Out}_c^{t1}\right) \\ \mathrm{Out}_c^{t3} = \mathrm{CAT}\left(\mathrm{Out}_c^{t2}, I_c^t\right) \\ \mathrm{Out}_c^{t} = \mathrm{Flt}\left(\mathrm{Conv}_{2\mathrm{D}}\left(\mathrm{Out}_c^{t3}\right)\right) \end{cases} \quad (4)$$

where Out_c^{t1} and Out_c^{t2} are the intermediate outputs of residual connection, $\mathrm{LN}(\cdot)$ is the layer normalization, Out_c^{t3} denotes the output after integrating the local feature with the class-wise correlation feature sequence, $\mathrm{Conv}_{2\mathrm{D}}$ is a 2-D convolution layer, Flt is the flatten operation, and Out_c^t is the integrated feature sequence of the cth class.

In DI2M, the integration of CNN with the transformer allows the module to focus on both patch-wise local information and class-wise long-range dependency information. CNN captures patch-wise information that can focus on the local personality characteristics of different samples, and then the transformer captures the relationship between various classes of support set samples and query set samples, which helps to better utilize the long-range information to learn class-wise correlation features while ignoring irrelevant information. Moreover, this integration can inject the inductive bias of CNN into the transformer, so that the transformer can perform more ideally under few-shot conditions, as well as making the mined sample features more discriminative and class-wise relevant. Ultimately, DI2M maps features to an embedding space where semantically similar features can be distributed closer together based on the capture of class-wise correlations between the support set and query set, which is conducive to better performing FSL.

C. Cross-Scene FSL

We accomplish cross-scene FSL by performing FSL using data from different domains during each iteration of training. Specifically, taking the source domain FSL as an example. DI2M extracts the integrated features and maps them into the embedding space, and FSL is carried out by measuring the distance between the features of the support set samples and query set samples, which are in the embedding space. First, calculate a corresponding prototype representation for different classes of support set samples

$$\mathrm{Pro}_c^{ss} = \frac{1}{\left|\mathrm{Out}_c^{ss}\right|} \sum_{\left(\mathrm{out}_{cl}^{ss}, y_{cl}\right) \in \mathrm{Out}_c^{ss}} \mathrm{out}_{cl}^{ss} \quad (5)$$

where Pro_c^{ss} is the prototype representation of the cth class sample in the support set, Out_c^{ss} is the set of integrated feature of the cth class samples in the support set, out_{cl}^{ss} denotes the lth integrated feature in the set, and y_{cl} is the corresponding label.

Subsequently, the distances between the query set samples and these prototype representations are measured by the Euclidean distance, and the probability distribution of the query set samples is obtained by Softmax

$$P\left(y = c \middle| \mathrm{out}_u^{\mathrm{sq}} \in \mathrm{Out}^{\mathrm{sq}}\right) = \frac{\exp\left(-Eu\left(\mathrm{out}_u^{\mathrm{sq}}, \mathrm{Pro}_c^{ss}\right)\right)}{\sum_{c=1}^{C} \exp\left(-Eu\left(\mathrm{out}_u^{\mathrm{sq}}, \mathrm{Pro}_c^{ss}\right)\right)} \quad (6)$$

where $\mathrm{Out}^{\mathrm{sq}}$ is the set of integrated feature of all samples in the source domain query set, $\mathrm{out}_u^{\mathrm{sq}}$ denotes the uth integrated feature in the set, and $Eu(\cdot, \cdot)$ is the Euclidean distance metric.

Similarly, the target domain FSL is executed to obtain the probability distribution of query set samples of the target domain

$$P\left(y = c \middle| \mathrm{out}_u^{\mathrm{tq}} \in \mathrm{Out}^{\mathrm{tq}}\right) = \frac{\exp\left(-Eu\left(\mathrm{out}_u^{\mathrm{tq}}, \mathrm{Pro}_c^{\mathrm{ts}}\right)\right)}{\sum_{c=1}^{C} \exp\left(-Eu\left(\mathrm{out}_u^{\mathrm{tq}}, \mathrm{Pro}_c^{\mathrm{ts}}\right)\right)} \quad (7)$$

where $\mathrm{Pro}_c^{\mathrm{ts}}$ is the prototype representation of the cth class sample of the target domain support set, $\mathrm{Out}^{\mathrm{tq}}$ is the set of integrated feature of all samples in the target domain query set, and $\mathrm{out}_u^{\mathrm{tq}}$ denotes the uth integrated feature in the set.

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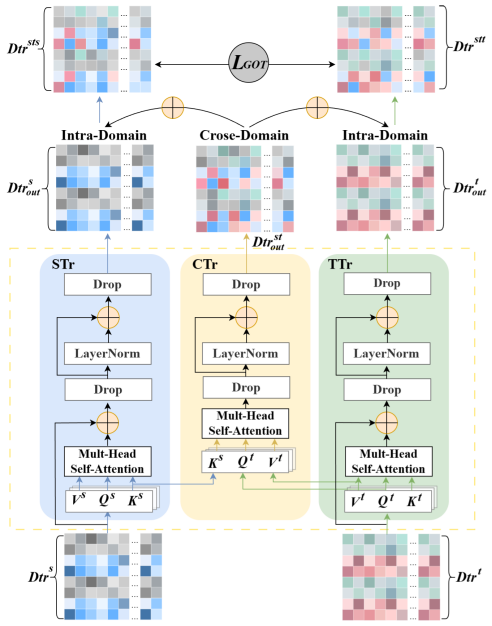

Fig. 3. Interdomain joint IDM.

Perform source domain FSL and target domain FSL alternately during the training iterations, i.e., performing distance metrics between features, which can optimize the feature embedding space so that features of the same class are close to each other while features of different classes are far away from each other. It enables the model to not only utilize the rich labeled data in VHRI to learn the changed information but also learn those specific changed features in the HSI scenario. The effective migration of learnable information from the source domain is realized, and the need for labeled data of HSI for model training is reduced.

D. IDM With Interdomain Collaboration

To alleviate the implications for knowledge learning and migration due to the intrinsic feature distribution differences of cross-scene data, an IDM is designed to capture the features of the two cross-scene domains and perform feature distribution alignment so that the distributions of the two domains gradually tend to be consistent to achieve domain adaptation. Fig. 3 illustrates the structure of IDM. It consists of three transformer-encoder-based variant structures, which are named source-transformer (STr), cross-transformer (CTr), and target-transformer (TTr). STr and TTr are used to extract intradomain discriminative features in the source and target domains, respectively, and CTr is used to extract interdomain commonality features.

1) Feature Data Processing: Perform the concatenation operation on all the integrated features of the source domain extracted by DI2M

$$Dtr^{s} = \mathrm{CAT}\left(\mathrm{Out}_1^{ss}, \ldots, \mathrm{Out}_c^{ss}, \mathrm{Out}_1^{sq}, \ldots, \mathrm{Out}_c^{sq}\right) \quad (8)$$

where Out_c^{ss} is the integrated feature set of the cth class sample of the source domain support set, and Out_c^{sq} is the integrated feature set of the cth class sample of the source domain query set.

Similarly, execute the same process for the target domain integrated features

$$Dtr^{t} = \mathrm{CAT}\left(\mathrm{Out}_1^{ts}, \ldots, \mathrm{Out}_c^{ts}, \mathrm{Out}_1^{tq}, \ldots, \mathrm{Out}_c^{tq}\right) \quad (9)$$

where Out_c^{ts} is the integrated feature set of the cth class sample of the target domain support set, and Out_c^{tq} is the integrated feature set of the cth class sample of the target domain query set.

2) Intradomain Discriminative Feature Extraction: In the two branches STr and TTr, intradomain discriminative features of the source and target domains are captured. Through the multihead self-attention layer, it can capture the intradomain personality feature while taking into account domain-wise long-range dependencies. First, the domain-wise feature sequences Dtr^{s} and Dtr^{t} are inputted into two branches, and processed by linear mapping and mapping function to obtain queries, keys and values, denoted as Q^{s}, K^{s}, V^{s}, Q^{t}, K^{t}, and V^{t}, where Q^{s} denotes the query obtained by Dtr^{s} and the rest is the same. Next, the dot product operation and Softmax function are applied between Q^{s} and K^{s} to obtain the domain-wise feature correlation score matrix for the source domain. Similarly, the correlation score matrix for the target domain can be obtained. Subsequently, they are dot produced with V^{s} and V^{t}, respectively. Finally, the dimensions of the matrix are rearranged and linearly mapped to obtain the feature maps Dtr^{s}_{out} for the source domain samples and Dtr^{t}_{out} for the target domain samples.

3) InterdDomain Joint Information Extraction: The extraction of interdomain joint information is implemented through CTr. Specifically, its key is composed of K^{s} obtained from the source domain, and its query and value are composed of Q^{t} and V^{t} obtained from the target domain, respectively. Through multilayer processing, the interdomain commonality feature map $Dtr^{\mathrm{st}}_{\mathrm{out}}$ is finally extracted

$$\begin{cases} Dtr^{\mathrm{st0}}_{\mathrm{out}} = \mathrm{Drop}\left(\mathrm{Lin}\left(\left(\mathrm{Softmax}\left(\dfrac{Q^{t}(K^{s})^{T}}{\sqrt{d^{\mathrm{st}}}}\right)\right)V^{t}\right)\right) \\ Dtr^{\mathrm{st1}}_{\mathrm{out}} = \mathrm{Drop}\left(Dtr^{\mathrm{st0}}_{\mathrm{out}}\right) \\ Dtr^{\mathrm{st}}_{\mathrm{out}} = \mathrm{Drop}\left(\mathrm{LN}\left(Dtr^{\mathrm{st1}}_{\mathrm{out}}\right) + Dtr^{\mathrm{st1}}_{\mathrm{out}}\right) \end{cases} \quad (10)$$

where $Dtr^{\mathrm{st0}}_{\mathrm{out}}$ and $Dtr^{\mathrm{st1}}_{\mathrm{out}}$ are the intermediate outputs of the residual connection operation for each layer, d^{st} denotes the feature dimension of K^{s}, and $(K^{s})^{T}$ is the transpose matrix of K^{s}.

4) Joint Information Allocation and Cross-Scene Feature Distribution Alignment: Allocate $Dtr^{\mathrm{st}}_{\mathrm{out}}$ to Dtr^{s}_{out} and Dtr^{t}_{out}

$$\begin{cases} Dtr^{\mathrm{sts}} = Dtr^{\mathrm{st}}_{\mathrm{out}} + Dtr^{s}_{\mathrm{out}} \\ Dtr^{\mathrm{stt}} = Dtr^{\mathrm{st}}_{\mathrm{out}} + Dtr^{t}_{\mathrm{out}} \end{cases} \quad (11)$$

where Dtr^{sts} is the distribution feature of the source domain after joint information allocation, and Dtr^{stt} is the distribution feature of the target domain after joint information allocation.

In IDM, the extraction of intradomain features in the source and target domains can help to capture domain-wise discriminative features, while the extraction of interdomain

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joint features enables the model to focus on the commonality features of the two domains. Allocating commonality feature information into discriminative features and combining them can better bridge the distributional differences between domains, improve the robustness of the model, and alleviate the problem of overfitting.

Finally, the alignment of Dtr^{sts} and Dtr^{stt} is achieved by graph optimal transport (GOT) [47]. Two key metric functions are included in GOT, Wasserstein distance (WD) and Gromov-WD (GWD). GOT is a useful tool that includes two metric functions, namely WD and GWD. These functions can be utilized to measure and minimize the dissimilarity between two distributions. As a result, GOT is commonly employed as a loss function for aligning feature distributions. First, the feature vector corresponding to each sample in the obtained Dtr^{sts} and Dtr^{stt} is regarded as a node, and the WD measures the distance between nodes in two domains, while GWD measures the distance between a pair of nodes in one domain and a pair of nodes in another domain. In GOT, combining the two metrics, WD and GWD, can be more effective in capturing and minimizing the differences between feature distributions. Therefore, the domain adaptation loss function based on GOT is as follows:

$$L_{\mathrm{GOT}} = \mathrm{GOT}\left(Dtr^{\mathrm{sts}}, Dtr^{\mathrm{stt}}\right) = \sum_{e,\hat{e},v,\hat{v}} \left(\mathrm{T}_{e,v} \cdot cs\left(dtr_e^{\mathrm{sts}}, dtr_v^{\mathrm{stt}}\right)\right) + \mathrm{T}_{\hat{e},\hat{v}} \cdot L\left(dtr_e^{\mathrm{sts}}, dtr_v^{\mathrm{stt}}, \hat{dtr}_e^{\mathrm{sts}}, \hat{dtr}_v^{\mathrm{stt}}\right) \quad (12)$$

where L_{GOT} is the GOT-based domain adaptation loss function. $cs(\cdot,\cdot)$ is the WD distance metric function that measures the distance between dtr_e^{sts} and dtr_v^{stt}. dtr_e^{sts} is the eth node of Dtr^{sts} and dtr_v^{stt} is the vth node of Dtr^{stt}. $L(\cdot,\cdot,\cdot,\cdot)$ is the GWD metric function that computes the dissimilarity between the node pair $(dtr_e^{\mathrm{sts}}, \hat{dtr}_e^{\mathrm{sts}})$ in the Dtr^{sts} and the node pair $(dtr_v^{\mathrm{stt}}, \hat{dtr}_v^{\mathrm{stt}})$ in the Dtr^{stt}. $T_{e,v}$ and $T_{\hat{e},\hat{v}}$ are the shared matrices of WD and GWD, and the process of solving them is to minimize the WD and GWD to reduce the difference in feature distribution to achieve domain adaptation.

E. Total Loss Function

After the above three processes of feature extraction of dual-information integrated mapping, cross-scene FSL, and domain adaptation with interdomain joint information allocation, CSIDC-FSL completes a training. We use the FSL and GOT loss functions to update the network parameters. In this case, the loss function of the source domain FSL can be obtained by calculating the query set sample probability distribution [obtained from (6)] and its corresponding true value

$$L_{\mathrm{FSL}}^{s} = -\log P\left(y = c \middle| \mathrm{out}_u^{\mathrm{sq}} \in \mathrm{Out}^{\mathrm{sq}}\right) = Eu\left(\mathrm{out}_u^{\mathrm{sq}}, \mathrm{Pro}_c^{ss}\right) + \log \sum_{c=1}^{C} \exp\left(-Eu\left(\mathrm{out}_u^{\mathrm{sq}}, \mathrm{Pro}_c^{ss}\right)\right). \quad (13)$$

In the same way, through (7), the target domain FSL loss function can be defined as follows:

$$L_{\mathrm{FSL}}^{t} = -\log P\left(y = c \middle| \mathrm{out}_u^{\mathrm{tq}} \in \mathrm{Out}^{\mathrm{tq}}\right) = Eu\left(\mathrm{out}_u^{\mathrm{tq}}, \mathrm{Pro}_c^{\mathrm{ts}}\right) + \log \sum_{c=1}^{C} \exp\left(-Eu\left(\mathrm{out}_u^{\mathrm{tq}}, \mathrm{Pro}_c^{\mathrm{ts}}\right)\right). \quad (14)$$

Therefore, the final loss function of CSIDC-FSL is denoted as follows:

$$L_{\mathrm{ALL}} = \begin{cases} L_{\mathrm{FSL}}^{s} + \lambda L_{\mathrm{GOT}}, & \text{episode } \% \ 2 = 0 \\ L_{\mathrm{FSL}}^{t} + \lambda L_{\mathrm{GOT}}, & \text{else} \end{cases} \quad (15)$$

where episode is the number of iterations rounds. λ is a hyperparameter, and its value will be discussed in Section V-A.

F. Testing Process

After DI2M is trained, the extracted features can form a suitable distribution in the embedding space, so that samples of the same class are gathered together, while samples of different classes are dispersed from each other. The CD task can be viewed as a binary classification problem that aims to distinguish the changed class from the unchanged class. So during the testing process, we first pretrained the nearest neighbor (NN) classifier using the HSI labeled samples that participated in the training. With the DI2M model, features are effectively mapped into the embedding space, where features of the same class are close to each other. This distribution characteristic enables NN classifier to use proximity relationships to classify extracted features, thereby accurately achieving CD.

IV. Experiment and Result Analysis

A. Source and Target Datasets Along With Experimental Settings

The CSIDC-FSL proposed in this article is a cross-scene FSL framework for HSI-CD. To verify its effectiveness and robustness, we conducted experiments on four widely used public HSI-CD datasets, that is, using the Bay Area dataset [48], Santa Barbara dataset [48], Farmland dataset [49], and Hermiston dataset [48] as the target domain data. And we use a VHRI CD dataset, namely the LEVIR-CD dataset [50] as the source domain. Among them, we randomly select a pair of bitemporal images as the source domain data. Detailed information on the source and target domain datasets is given in Table I and Fig. 4.

In the experiment, CSIDC-FSL is implemented based on the PyTorch framework and the NVIDIA RTX 3080 Ti GPU. For the input data $\{S_s, Q_s, S_t, Q_t\} \in \mathbb{R}^{p\times p\times F}$, where p is set to 9, F is set to 100. The number of training iterations is 2000, and adaptive moment estimation (Adam) [51] is used as the optimizer with the learning rate tuned to 0.001. For the training of the model, 500 labeled samples per class are randomly selected from the source domain and five labeled samples per class are randomly selected from the target domain. In addition, for the sampling that divides the support set and query set in FSL, we select one sample per class in the training samples to constitute the support set and 19 samples per class to constitute the query set. In the comparison experiments, to ensure the fairness of comparison, all comparison methods use the same sampling rate as CSIDC-FSL.

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TABLE I
DATASETS INFORMATION

Dataset	Type	Domain	T1/T2	Size	Channels	Class		
						No.	Name	Number of samples
LEVIR-CD	VHRI	Source	2002/2013	1024*1024	3	1	changed	121949
						2	unchanged	926627
Bay Area	HSI	Target	2013/2015	600*500	224	1	changed	39270
						2	unchanged	34211
Santa Barbara	HSI	Target	2013/2014	984*740	224	1	changed	52134
						2	unchanged	80418
Farmland	HSI	Target	2006/2007	450*140	155	1	changed	18277
						2	unchanged	44723
Hermiston	HSI	Target	2004/2007	200*390	242	1	changed	9986
						2	unchanged	68014

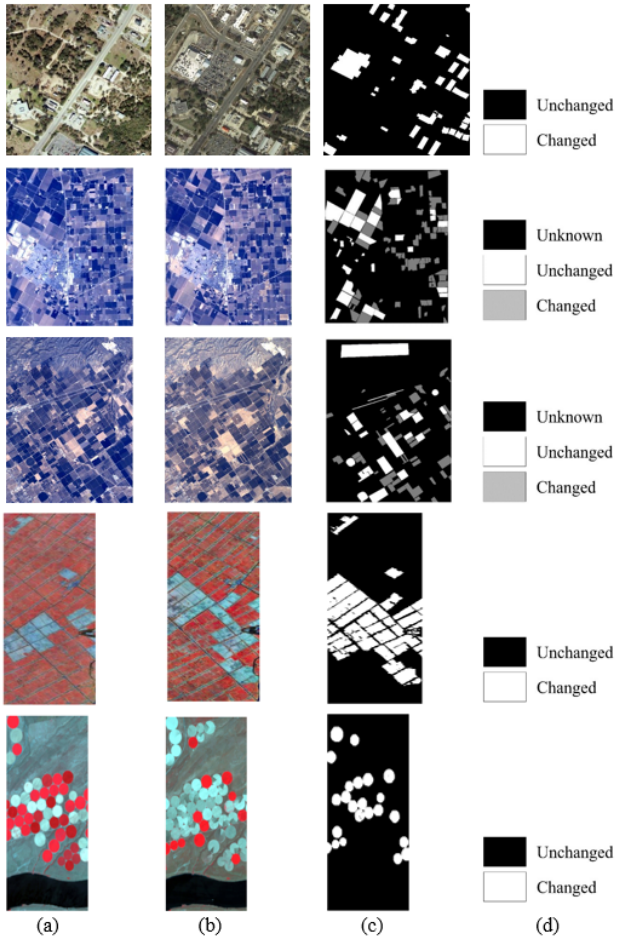

Fig. 4. (a) Image acquired on $T1$. (b) Image acquired on $T2$. (c) Ground-truth image. (d) Legend.

B. Introduction to Comparison Methods and Evaluation Indicators

1) Comparison Methods: In order to confirm the performance of the CSIDC-FSL in CD, it is compared with nine advanced methods in recent years. It is worth noting that, to ensure the fairness of the experiment, for the FGDANet method (an unsupervised method), we use the data in the original paper on the Bay Area and Santa Barbara datasets. We use the "-" symbol to indicate the missing evaluation indicator values. Details of these comparison methods are as follows.

CSANet (2022) [52]: A network that uses a dual-branch attention mechanism to effectively extract and combine spatial–spectral–temporal features to enhance the discriminative ability of changed features. However, the feature fusion process of the network is too simple, which easily affects the ability to capture details.

SSTFormer (2022) [40]: A network based on the transformer. It can achieve good results when trained with fewer samples. Nonetheless, the model's performance declines when confronted with an exceedingly limited number of training samples.

DAFormer (2023) [53]: A semisupervised method based on a dual-branch transformer structure, aiming to overcome the challenge of limited training samples through a semisupervised domain alignment model. However, when the sample size is extremely small, the accuracy of the model needs to be further improved.

DCFSL (2022) [32]: A network based on cross-domain FSL, aiming to reduce the dependence of model training on the number of samples through FSL. However, the network can only focus on local features and fails to fully consider the capture and integration of long-range dependency information, thereby constraining its proficiency in recognizing complex patterns to some degree.

Gia-CFSL (2024) [54]: A network based on the FSL strategy, which realizes the capture and transfer of nonlocal spatial information on the data of cross scene while performing FSL, effectively improving the performance under the few-shot condition. However, the robustness of the model needs to be further improved.

MSDFFN (2023) [55]: A network that guides multiscale feature extraction and fusion through attention, which can better represent subtle changed features. However, this network may face challenges when dealing with the few-shot problem.

BiG-FSLF (2023) [56]: A network that employs the FSL strategy, which extracts features from heterogeneous domain data and transfers and utilizes knowledge via FSL, and is better equipped to handle the challenges posed by few-shot scenarios. However, it cannot effectively model long-range information, which limits the ability of the model to understand changes in image scenes.

FGDANet (2023) [23]: An unsupervised Siamese network. It designs a domain adaptation model to promote effective learning of cross-domain information. However, the

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TABLE II
EXPERIMENTAL RESULTS ON BAY AREA AND SANTA BARBARA DATASETS

Methods	Bay Area					Santa Barbara				
	OA	AA	Kappa	P	F1	OA	AA	Kappa	P	F1
CSANet	0.7090	0.7410	0.4292	0.8557	0.7352	0.8308	0.8503	0.6115	0.8810	0.8706
SSTFormer	0.7192	0.7106	0.4281	0.6994	0.7572	0.7799	0.7817	0.5497	0.7024	0.7368
DAFormer	0.7038	0.7427	0.3962	0.8162	0.7563	0.7639	0.7595	0.5065	0.7861	0.7093
DCFSL	0.8627	0.8651	0.7261	0.9108	0.8638	0.9161	0.9052	0.8218	0.9280	0.8891
Gia-CFSL	0.8959	0.8954	0.7906	0.9067	0.9027	0.9206	0.9151	0.8332	0.9098	0.8981
MSDFFN	0.7808	0.7859	0.5656	0.8576	0.7699	0.8349	0.8181	0.6470	0.8251	0.7771
BiG-FSLF	0.8935	0.8974	0.7890	0.9561	0.9012	0.9209	0.9108	0.8320	0.9326	0.8955
FGDANet	0.9022	-	0.7899	-	-	0.9023	-	0.8012	-	-
CSIDC-FSL	**0.9331**	**0.9347**	**0.8661**	**0.9612**	**0.9354**	**0.9312**	**0.9227**	**0.8542**	**0.9385**	**0.9098**

performance of FGDANet largely depends on the quality of pseudo labels.

DCENet (2024) [57]: A semisupervised HSI-CD network that works on combining labeled and unlabeled samples to train the network. The performance is better in the case of small-sample-size sampling rates, but there is a need to improve the model's performance under the few-shot condition.

2) Evaluation Indicators: In order to reduce the randomness of the results and improve reliability, each experiment is repeated five times, and the average of the results of these five experimental results is used as the final evaluation result. When evaluating CD performance, we use several key evaluation indicators: overall accuracy (OA), average accuracy (AA), Kappa coefficient (Kappa), precision (P), and $F1$-score ($F1$). OA is the most intuitive performance evaluation indicator. It calculates the proportion of correctly classified samples in all categories to the total number of samples. AA calculates the accuracy for each category separately and then takes the average value to reflect the average performance of the model on all categories. Kappa is used to measure the global consistency of the predicted classification of the model. P is used to measure the precision of positive examples (changed class), while F1 further averages the P and recall. The higher the values of these evaluation indicators, the better the CD performance of the model. These indicators are calculated as follows:

$$\begin{cases} \mathrm{OA} = \dfrac{\mathrm{TP}+\mathrm{TN}}{\mathrm{TP}+\mathrm{TN}+\mathrm{FP}+\mathrm{FN}} \\ \mathrm{AA} = \dfrac{\sum_{i=1}^{n} \mathrm{accuracy}_i}{n} \\ \mathrm{PE} = \dfrac{(\mathrm{TP}+\mathrm{FP})\times(\mathrm{TP}+\mathrm{FN})+(\mathrm{TN}+\mathrm{FP})\times(\mathrm{TN}+\mathrm{FN})}{(\mathrm{TP}+\mathrm{TN}+\mathrm{FP}+\mathrm{FN})^2} \\ \mathrm{Kappa} = \dfrac{\mathrm{OA}-\mathrm{PE}}{1-\mathrm{PE}} \\ P = \dfrac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}} \\ F1 = 2\times\dfrac{P\times\frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}}}{P+\frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}}} \end{cases} \tag{16}$$

where P and N in TP, TN, FP, and FN, respectively, indicate that the sample predicted by the model is changed or unchanged, and T and F indicate that the model prediction is correct or wrong. n denotes the number of classifications. $\mathrm{accuracy}_i$ is the classification accuracy of the ith class.

C. Comparison of Experimental Results and Analysis

We give the experimental results in Tables II–IV. Based on the values of these evaluation indicators, CSIDC-FSL has achieved optimal results, which confirms that CSIDC-FSL has high CD detection performance when training samples are limited. In addition, it can be seen from the stable performance of CSIDC-FSL on the four datasets that it also has good robustness.

1) Results of Experiments on the Bay Area Dataset: On the Bay Area dataset, three frameworks based on cross-domain FSL, DCFSL, Gia-CFSL, and BiG-FSLF, have all achieved good results in quantitative evaluation metrics, with CSIDC-FSL achieving the best performance. Compared with other supervised methods, the above results significantly validate the advantages of FSL in utilizing limited samples for effective learning. For unsupervised FGDANet, its performance also confirms that unsupervised learning still has certain effectiveness in handling few-shot situations. However, its Kappa still has a gap compared with the results of FSL-based methods. This gap may indicate that, in the absence of real labels during training, the model is prone to bias when learning changed features. In addition, we also present the prediction result images and ground-truth image of all methods on the Bay Area dataset in Fig. 5. Based on the images, it is quite evident that there is a considerable difference among the prediction results of CSANet, SSTFormer, DAFormer, and MSDFFN compared to the ground-truth image. This deviation is most significant in the part within the blue box in the image. Therefore, it can be concluded that these methods are difficult to show good performance when there are very few training samples. As can be seen from the red frame of the prediction images, this seems to be an area that is difficult to predict accurately. Most methods have large areas of false detection in this region. In contrast, the method proposed in this article can make better predictions in this area.

2) Results of Experiments on the Santa Barbara Dataset: On the Santa Barbara dataset, the FSL-based method and the unsupervised-based method can achieve better results. CSANet and MSDFFN can achieve good results under conventional sampling, but their detection performance decreases under the few-shot condition. Although SSTFormer and DAFormer can achieve ideal results with fewer training samples, they are

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TABLE III
EXPERIMENTAL RESULTS ON FARMLAND AND HERMISTON DATASETS

Methods	Farmland					Hermiston				
	OA	AA	Kappa	P	F1	OA	AA	Kappa	P	F1
CSANet	0.9012	0.8980	0.7382	0.6245	0.8146	0.9446	0.8799	0.7445	0.7708	0.8159
SSTFormer	0.9207	0.9393	0.8204	0.7935	0.8782	0.8768	0.8853	0.6076	0.5698	0.6723
DAFormer	0.8925	0.8766	0.7363	0.7922	0.8111	0.8704	0.7479	0.5517	0.8030	0.6220
DCFSL	0.9281	0.9171	0.8268	0.8662	0.8777	0.9633	0.9014	0.8273	0.8910	0.8480
Gia-CFSL	0.9316	0.9346	0.8395	0.8467	0.8885	0.9557	0.9324	0.8171	0.8028	0.8423
MSDFFN	0.9009	0.8824	0.7566	0.8247	0.8252	0.9427	0.9224	0.7690	0.7322	0.8017
BiG-FSLF	0.9353	0.9338	0.8469	0.8598	0.8931	0.9507	0.9188	0.7998	0.8024	0.8278
FGDANet	0.9340	0.9375	0.8452	0.8451	0.8926	0.8803	0.8863	0.5903	0.5188	0.6567
CSIDC-FSL	**0.9415**	**0.9416**	**0.8615**	**0.8679**	**0.9034**	**0.9743**	**0.9461**	**0.8854**	**0.8933**	**0.9006**

TABLE IV
RESULTS OF EXPERIMENTS UNDER THE SMALL-SAMPLE CONDITION

Datasets	Methods	Samples	OA	Kappa
Bay Area	DAFormer	200	97.12	94.21
	SSTFormer	200	95.87	91.72
	DCENet	200	96.33	92.52
	CSIDC-FSL	200	**97.19**	**94.35**
Santa Barbara	DAFormer	100	94.55	87.10
	SSTFormer	100	95.56	90.76
	DCENet	93	94.21	87.81
	CSIDC-FSL	80	**95.75**	**90.98**
Hermiston	DAFormer	200	98.26	92.38
	SSTFormer	200	98.12	91.71
	DCENet	116	98.22	91.91
	CSIDC-FSL	60	**98.28**	**92.44**

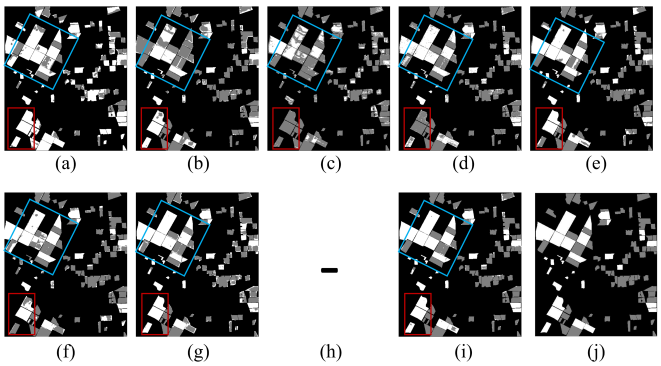

Fig. 5. Bay Area. (a) CSANet. (b) SSTFormer. (c) DAFormer. (d) DCFSL. (e) Gia-CFSL. (f) MSDFFN. (g) BiG-FSLF. (h) FGDANet. (i) CSIDC-FSL. (j) Ground truth.

difficult to achieve good results with extremely few training samples. Overall, CSIDC-FSL has achieved the best level in all quantitative evaluation indicators, especially in Kappa and $F1$, where it has made significant progress. Compared to the suboptimal results, Kappa improved by 2.1% and $F1$ improved by 1.17%. The CD result image and ground-truth image of each method on the Santa Barbara dataset are shown in Fig. 6. By observing the surface prediction results within the blue and red borders, it is easy to see that the prediction results of CSIDC-FSL are closest to the ground-truth image, and its prediction accuracy is better than other methods.

3) Results of Experiments on the Farmland Dataset: Fig. 7 shows the prediction result images corresponding to different methods. As can be seen from the ground-truth image, the changed and unchanged areas of Farmland are quite confused, and the class boundaries are blurred and discontinuous. In this

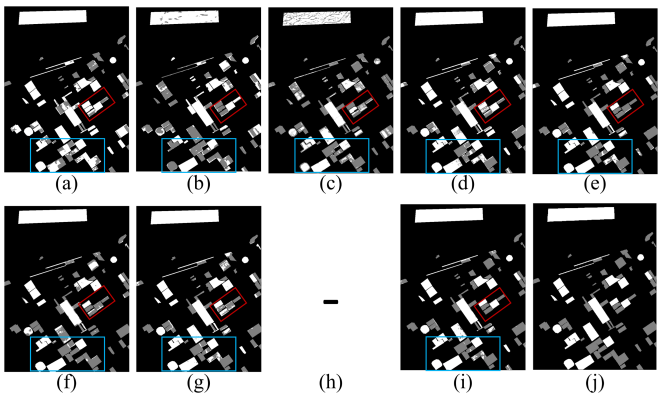

Fig. 6. Santa Barbara. (a) CSANet. (b) SSTFormer. (c) DAFormer. (d) DCFSL. (e) Gia-CFSL. (f) MSDFFN. (g) BiG-FSLF. (h) FGDANet. (i) CSIDC-FSL. (j) Ground truth.

case, the CSANet, Gia-CFSL, MSDFFN, and SSTFormer methods are not ideal for determining the boundaries of the changed region and are not accurate enough for predicting the changed and unchanged classes. Methods such as DAFormer and FGDANet have errors in the prediction of large unchanged areas, resulting in noise-like interference in the predicted images. In addition, in the area marked by the red frame line in the predicted image, all comparison methods failed to make accurate judgments. In contrast, the prediction result of CSIDC-FSL is most consistent with the ground-truth image, which highlights the advantages of the method proposed in this article in effectively predicting changes.

4) Results of Experiments on the Hermiston Dataset: Comprehensively analyzing the data in Tables II and III, we can observe that under the condition of few-shot, CSANet, which is based entirely on CNN, shows better performance than SSTFormer and DAFormer, which are based on transformer. This phenomenon is particularly prominent on the Santa Barbara dataset. This shows that since transformer lacks the inductive bias inherent in CNN, its feature learning ability is not as powerful as CNN when the training samples are extremely limited. This finding further verifies that the design idea of CSIDC-FSL combining CNN and transformer is reasonable. The designed integrated structure can take into account the advantages of both networks, thereby achieving more efficient feature extraction and learning under the few-shot condition. In Fig. 8, we can see the comparison of the prediction results of various methods with the ground-truth image. Obviously,

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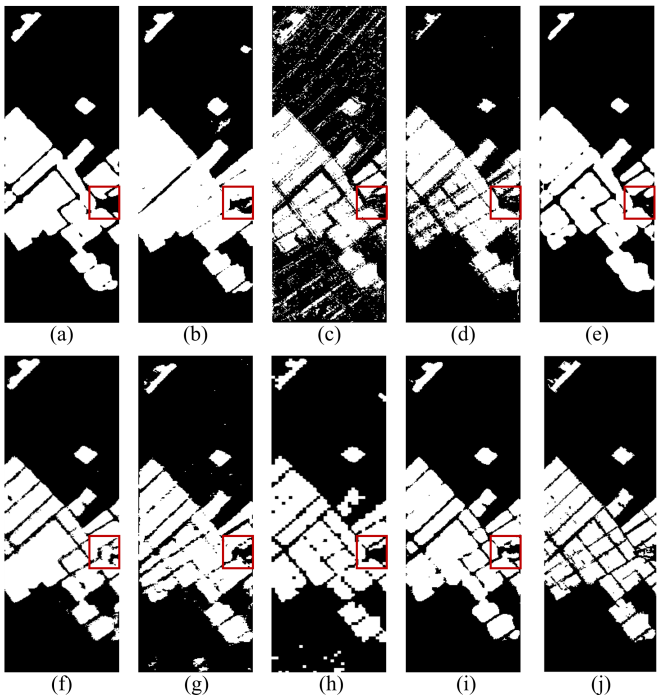

Fig. 7. Farmland. (a) CSANet. (b) SSTFormer. (c) DAFormer. (d) DCFSL. (e) Gia-CFSL. (f) MSDFFN. (g) BiG-FSLF. (h) FGDANet. (i) CSIDC-FSL. (j) Ground truth.

BiG-FSLF, MSDFFN, and FGDANet produce more false detections in prediction, while DCFSL shows some missed detection problems. In contrast, CSIDC-FSL not only visually matches the ground-truth image the best, but also reaches the best level in terms of quantitative evaluation indicators.

5) Results of Experiments Under the Small-Sample Condition: When comparing different methods, we notice that some methods used for comparison are based on small-sample-size sampling, usually around 100–200 samples from each class are selected for training. To ensure the fairness of the comparison and enhance the comprehensiveness of the experiment, we also conduct a series of experiments under the small-sample condition. The experimental results are given in Table IV. For the Bay Area dataset, all methods are performed under the same conditions using 200 training samples. Under the small-sample condition, the results achieved by the proposed CSIDC-FSL are still competitive. On the Santa Barbara dataset, CSIDC-FSL only uses 80 training samples for experiments. Even though the number of training samples is smaller than other comparison methods, CSIDC-FSL has a 0.22% improvement in Kappa compared with SSTFormer. When conducting experiments on the Santa Barbara dataset, DCENet selects 93 samples for training. This selection is based on the data settings in the original paper. For the Hermiston dataset, the DCENet can still achieve better detection results than the other two comparison methods even if it uses fewer training samples. This phenomenon may be attributed to DCENet's more efficient feature extraction capabilities and its strategy of combining labeled and unlabeled information for joint training. The proposed CSIDC-FSL uses the fewest training samples in the experiment but still performs best. This result

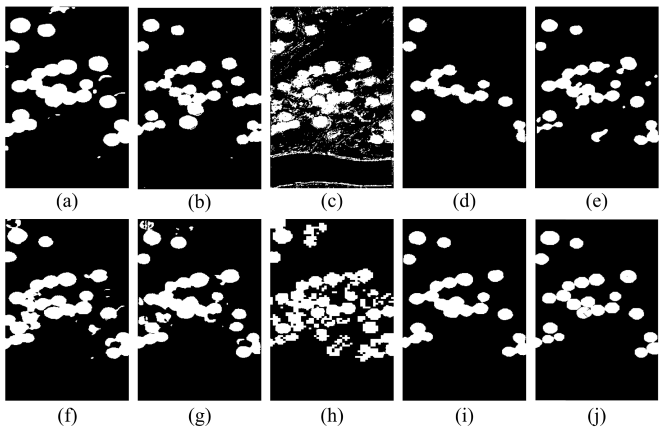

Fig. 8. Hermiston. (a) CSANet. (b) SSTFormer. (c) DAFormer. (d) DCFSL. (e) Gia-CFSL. (f) MSDFFN. (g) BiG-FSLF. (h) FGDANet. (i) CSIDC-FSL. (j) Ground truth.

reflects that CSIDC-FSL not only performs well under the few-shot condition but also performs well under the small-sample condition. This excellent performance may stem from the unique multilevel feature hybrid perceptual mechanism of CSIDC-FSL and its ability to effectively utilize learnable information in the existing remote sensing data for deep mining and learning.

D. Ablation Experimental Results and Analysis

CSIDC-FSL consists of three main components: the DI2M for feature extraction, the cross-scene FSL, and the IDM for capturing domain-wise distribution features for domain adaptation. In Table V, we give the results of ablation experiments for each module to confirm the necessity and effectiveness of each part in CSIDC-FSL. We design five ablation models, which are the following: 1) "Baseline:" only use the 3-D convolution layers in DI2M to extract features, using this most basic structure as baseline; 2) "W-DI2M:" remove cross-scene FSL and IDM, which only uses DI2M to extract features; 3) "WO-DA:" remove IDM and GOT loss from CSIDC-FSL, which does not implement domain adaptation for cross-scene tasks; 4) "WO-FSL:" remove cross-scene FSL; and 5) "WO-CLAT:" remove CLAT from CSIDC-FSL. The experiment is repeated five times and the average value is taken as the ablation result.

Table V "a" shows the results of using 3-D convolution layers to extract local spatial–spectral features for CD. From the evaluation indices values obtained on the four datasets, it can be concluded that the simple CNN network structure focuses on the local information of the samples and fails to extract the deep intrinsic features well, resulting in an unsatisfactory CD accuracy. On the other hand, Table V "b" reveals that using DI2M to focus on both local-wise and class-wise long-range information yields better results than the ablation model "a," which reflects that the DI2M is more effective in capturing the changed characteristics. Table V "e" conducts experiments by removing the CLAT module from the entire framework to further confirm the importance of focusing on local and class-wise long-range information. Due to the lack of CLAT, the model cannot capture class-wise long-range information and integrate it with local-wise information.

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TABLE V
ABLATION EXPERIMENTAL RESULTS

	a	b	c	d	e	f
Model	Base -line	W -DI2M	WO -DA	WO -FSL	WO -CLAT	CSIDC -FSL
Dataset	Bay Area					
OA(%)	86.14	87.76	91.81	90.07	89.37	**93.31**
Kappa(%)	71.93	75.43	83.59	79.92	78.69	**86.61**
Dataset	Santa Barbara					
OA(%)	84.37	88.20	91.97	90.66	90.88	**93.12**
Kappa(%)	68.49	75.32	83.08	80.94	80.68	**85.42**
Dataset	Farmland					
OA(%)	88.22	91.89	92.85	92.57	93.84	**94.15**
Kappa(%)	74.42	81.27	83.09	83.01	85.14	**86.15**
Dataset	Hermiston					
OA(%)	95.96	96.05	96.77	96.52	97.09	**97.43**
Kappa(%)	82.21	82.87	85.32	85.74	86.14	**88.54**

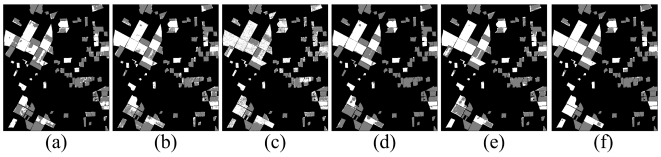

Fig. 9. Ablation experiment result images of the Bay Area dataset. (a) Baseline. (b) W-DI2M. (c) WO-DA. (d) WO-FSL. (e) WO-CLAT. (f) CSIDC-FSL.

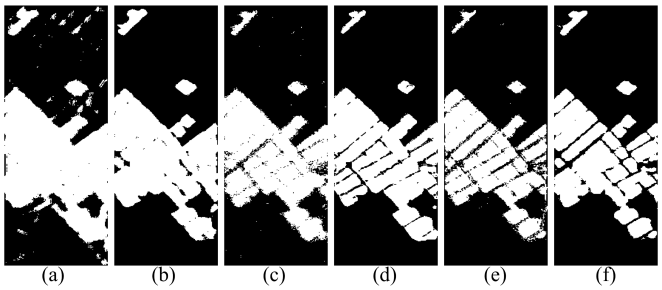

Fig. 10. Ablation experiment result images of the Farmland dataset. (a) Baseline. (b) W-DI2M. (c) WO-DA. (d) WO-FSL. (e) WO-CLAT. (f) CSIDC-FSL.

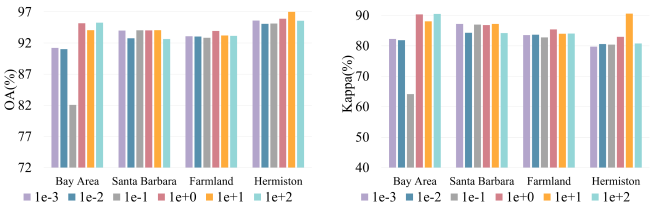

Fig. 11. Hyperparameter values on the four datasets. (a) OA obtained by CSIDC-FSL based on different values of λ. (b) Kappa obtained by CSIDC-FSL based on different values of λ.

Compared with the experimental results of the complete framework, the performance of the version without CLAT has declined. This result reflects the important role of class-wise long-range information in improving model performance. It highlights the necessity of integrating local-wise information with class-wise long-range information.

Table V "c" shows the experimental results of removing the domain adaptation process from CSIDC-FSL. It is evident that without domain adaptation, the accuracy of the CD will decrease for data from two domains, which are cross-scene and have differences in feature distributions. To effectively learn and transfer cross-scene knowledge, it is crucial to capture domain-wise distribution features using IDM and align these features so that the distributions of the two domains can be collaboratively consistent. Furthermore, as can be seen from the results without cross-scene FSL in Table V "d" that cross-scene FSL is equally important for CD under the few-shot conditions.

Overall, comparing the results of the different ablation models in each column, CSIDC-FSL produces the most favorable results. This can also reflect that DI2M, cross-scene FSL, and IDM are all integral parts of CSIDC-FSL that can achieve better CD results. In Figs. 9 and 10, we provide the prediction results of different ablation models on the Bay Area and Farmland datasets.

V. DISCUSSION

A. Hyperparameter Tuning

An adjustable regularization hyperparameter λ is added to (15). We discuss the impact of the value of λ on CSIDC-FSL in this section. We use five hyperparametric values for experiments and select appropriate hyperparametric values for different datasets according to the OA and Kappa values obtained from the experiments. Analyzing Fig. 11, it can be seen that it is not the case that the higher the value of λ, the better the performance of CSIDC-FSL, but its too small value also has a negative impact on the performance of the model. Overall, when λ is adjusted to $1e+2$, $1e+1$, $1e+0$, and $1e+1$, respectively, CSIDC-FSL can achieve better performance on the Bay Area, Santa Barbara, Farmland, and Hermiston datasets accordingly.

B. Feature Distribution Alignment Visualization

For cross-scene tasks, the source and target domain data will have different feature distributions due to different surface scenes, which is disadvantageous for cross-scene knowledge learning and migration. How to alleviate the differences in feature distributions is an issue that must be paid attention to. Because bridging the gap between the feature distributions of the two domains, i.e., domain adaptation, ensures that the model can more accurately capture and learn the shareable information and complete migration so that it can utilize the knowledge of the source domain data to assist the target domain in better realizing the CD. To this end, we design the IDM to focus on the intradomain personality features and interdomain commonality features, allocating the commonality features to the two domains and combining them with the personality features. Then the feature distributions are aligned through GOT to make the distributions of the two domains collaboratively consistent and achieve domain adaptation. To more intuitively show the effect of this process to align the feature distributions, we provide visual images of the feature distributions of the source and target domain data after T-SNE and normalization operation in Fig. 12, in which the red scatters are the source domain data and the green scatters are the target domain data. Fig. 12(a) shows the original feature distributions of the source and target domain data, from which it can be seen that, in the 2-D space with

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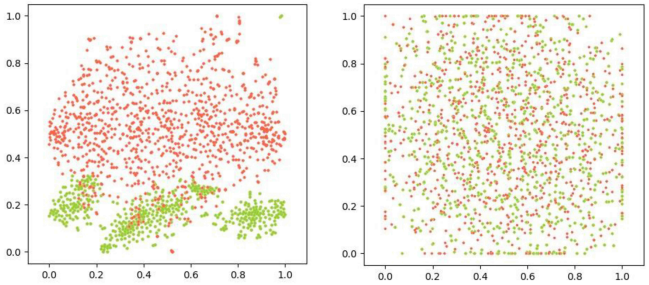
TABLE VI
COMPUTATIONAL COMPLEXITY COMPARISON OF DIFFERENT METHODS

Method		CSANet	SSTFormer	DAFormer	DCFSL	Gia-CFSL	MSDFFN	BiG-FSLF	FGDANet	CSIDC-FSL
Params(M)		2.39	2.54	0.41	4.26	0.31	38.53	0.66	2.02	1.70
Computing time(s)	Bay Area	249.17	139.06	168.74	381.91	325.43	423.51	454.76	1266.81	463.66
	Santa Barbara	449.65	335.73	352.49	515.26	464.22	432.33	473.74	1266.81	533.84
	Farmland	151.73	80.08	65.06	293.71	473.89	410.5	311.63	46.8918	314.03
	Hermiston	315.94	149.35	132.74	403.18	359.29	419.25	301.08	46.8918	395.62



(a) (b)

Fig. 12. Visualization result images by T-SNE. (a) Original feature distributions. (b) Feature distributions after domain adaptation.

horizontal and vertical coordinates of 0–1, the features of the source domain data are mostly distributed in the horizontal strip area of 0.4–0.8 in the upper middle part, while most of the features of the target domain data are concentrated in the lower half of the space. The distribution of the features of the original source and target domains is quite different and do not overlap very much. Fig. 12(b) is the visualization result after the implementation of feature distribution alignment, in which the difference between the feature distributions of the source domain and the target domain is significantly reduced and almost consistent, which confirms the effectiveness of the feature distribution alignment in CSIDC-FSL.

C. Computational Complexity Analysis

Computational complexity is also an important measure of the network. For comparison, we experiment with the same conditions, that is, the patch size is set to 9×9 and the batch size is set to 100. Finally, according to the experimental results, Table VI is obtained, including the number of parameters of different methods and their computing time (total time for training and testing) on the four datasets.

According to the data in Table VI, compared with other methods, the number of parameters of DAFormer, Gia-CFSL, and BiG-FSLF is relatively small. Among these methods, MSDFFN has the largest number of parameters. The number of parameters of our proposed CSIDC-FSL is at a medium level. According to the statistics of the overall computing time, FGDANet takes the longest time, which indicates that its computational complexity is relatively high. Even though BiG-FSLF has a small number of parameters, its computing time is relatively long. This can be attributed to the generative adversarial process that occurs within the network. On the other hand, SSTFormer has a relatively short computing time, but it requires a significant amount of time for data preprocessing in actual experiments. The CSIDC-FSL we propose is slightly longer in computing time, but the data preprocessing speed is fast. Moreover, it is worth noting that CSIDC-FSL achieves the best detection performance on four different datasets. Given its excellent performance, the calculation cost of this degree is reasonable and acceptable.

TABLE VII
ROBUSTNESS DISCUSSION

Source Domain	Target Domain				
	Farmland				
Hermiston	OA	AA	Kappa	P	F1
	0.9337	0.9115	0.8365	0.9083	0.8826
	Hermiston				
Farmland	OA	AA	Kappa	P	F1
	0.9726	0.9187	0.8719	0.9352	0.8876
	Bay Area				
Farmland	OA	AA	Kappa	P	F1
	0.9255	0.9285	0.8515	0.9725	0.9267

D. Robustness Discussion

To verify the robustness of the CSIDC-FSL, we design an additional set of experiments in which the source domain data are HSIs. The results of these experiments are shown in Table VII. Compared with the experimental results in Tables II and III, when using HSIs as source domain data, there is no particularly obvious difference in the performance we obtained, which demonstrates the robustness of the CSIDC-FSL. Especially when using the Farmland dataset as the source domain, the performance indicators obtained by CSIDC-FSL on the Bay Area dataset are still better than all comparison methods. However, compared with using the VHRI dataset as the source domain, the performance indicators have decreased, which may be attributed to the failure to re-explore and adjust the optimal hyperparameters in the new scenario.

VI. CONCLUSION

In this article, the CSIDC-FSL is proposed to provide a solution to the challenge of insufficient labeled samples for HSI-CD. CSIDC-FSL performs FSL on two cross-scene datasets to achieve effective migration of learnable information in the VHRI. Cross-scene FSL can utilize the information in the VHRI, where sample labeling is less costly, to assist HSI for CD while reducing the dependence of model training on the number of training samples. Specifically, it is realized through the following contributions: first, DI2M is proposed to capture patch-wise features and class-wise long-range correlations to mine more discriminative information. The designed integration structure can introduce suitable inductive bias for the transformer to perform well even with limited training samples. Then, IDM is proposed for intradomain distribution self-attention and interdomain distribution joint attention, aiming to focus on intradomain distribution characteristics

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and interdomain distribution commonality. Then the feature distributions are aligned to make the distributions collaboratively consistent and mitigate the impact of cross-scene feature distribution differences on FSL. Extensive experiments are conducted under few-shot and small-sample conditions. The results demonstrate that the proposed CSIDC-FSL exhibits superior CD detection performance at extremely low and moderately low sampling rates. It can effectively identify changed areas in datasets that are difficult to detect accurately by most other methods. However, the performance of CSIDC-FSL is not ideal when detecting some very subtle changes. Future research will be dedicated to designing more efficient feature extraction modules to further improve performance. In addition, based on parameter counts and the analysis of the total computing time for network training and testing, the current network architecture is not lightweight enough and its computational speed is not optimal. Future work will consider optimizing the model to reduce its complexity and increase processing speed.

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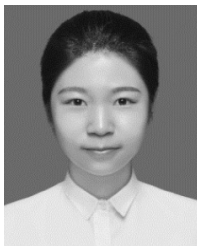

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